

WEEK 14

GEE Advanced: Landsat

Multi-Decadal Trend Analysis & Resilience

ARIA v9.5 -- The Resilience Monitor

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The Time Scale Gap

W13: Sentinel-2 (6 Years)

10m

High spatial resolution

6 yr

2020–2026 coverage

5 day

Revisit frequency

S2

Single-mission data

Q?

"What changed after the EQ?"

W14: Landsat (26 Years)

30m resolution, 4 missions merged

2000–2026 continuous record

Pixel-level linear trend analysis

Pond disappearance tracking

Vegetation resilience metrics

6

years (W13)



26

years (W14)

=

4x

longer record

+

4

missions

Landsat Mission History

1984–2012	1999–present	2013–present	2021–now
Landsat 5 TM Thematic Mapper SR_B1–B5, B7 30m resolution	Landsat 7 ETM+ Enhanced TM+ Same band numbers as L5 SLC failure May 2003	Landsat 8 OLI Operational Land Imager SR_B2–B7 (shifted!) Improved radiometry	Landsat 9 OLI-2 sensor Same bands as L8 Latest mission

W14 Uses 2000–2026 (26 Years)

L5 (2000–2012) + L7 (2000–present) + L8 (2013–present) + L9 (2021–present) merged into one continuous time series. Why not 1984? Landsat 5 pre-2000 data has quality gaps; starting at 2000 gives a cleaner baseline.

40+

years of archive

30m

resolution

8–16d

revisit per sat

Free

open access

Band Harmonization Challenge

Same spectral region, different band numbers -- NIR is B4 in L5/L7 but B5 in L8/L9!

Spectral	L5 TM	L7 ETM+	L8 OLI	L9 OLI-2	Common
Blue	SR_B1	SR_B1	SR_B2	SR_B2	Blue
Green	SR_B2	SR_B2	SR_B3	SR_B3	Green
Red	SR_B3	SR_B3	SR_B4	SR_B4	Red
NIR	SR_B4	SR_B4	SR_B5	SR_B5	NIR
SWIR1	SR_B5	SR_B5	SR_B6	SR_B6	SWIR1
SWIR2	SR_B7	SR_B7	SR_B7	SR_B7	SWIR2

Solution: Rename Function

Write a function that renames bands from each mission to a common set:

`rename_l57(image)` maps SR_B1→Blue, SR_B2→Green, SR_B3→Red, SR_B4→NIR, SR_B5→SWIR1, SR_B7→SWIR2

`rename_l89(image)` maps SR_B2→Blue, SR_B3→Green, SR_B4→Red, SR_B5→NIR, SR_B6→SWIR1, SR_B7→SWIR2

After renaming, `NDVI = normalizedDifference(['NIR', 'Red'])` works for ALL missions!

QA_PIXEL Cloud Masking

W13: Sentinel-2 SCL Band

Scene Classification Layer

Pixel values = land cover classes

scl.eq(4) = Vegetation

scl.eq(5) = Bare soil

Direct class comparison

W14: Landsat QA_PIXEL

Bitmask encoding

Each bit = a different flag

Bit 3 = Cloud

Bit 4 = Cloud Shadow

Use bitwiseAnd() to check

```
def mask_landsat_clouds(image):  
    qa = image.select('QA_PIXEL')  
    cloud_mask = qa.bitwiseAnd(1 << 3).eq(0)    # Bit 3 = cloud  
    shadow_mask = qa.bitwiseAnd(1 << 4).eq(0)    # Bit 4 = shadow  
    return image.updateMask(cloud_mask.And(shadow_mask))
```

Landsat 7 SLC-off Issue

The Problem (2003)



Scan Line Corrector (SLC) failed in May 2003. All Landsat 7 images after this date have wedge-shaped gaps (stripe artifacts) affecting ~22% of each scene.

The Solution: Median Compositing



When computing temporal composites (annual median), the stripes are in different positions across different orbits. Enough images = gaps fill automatically via pixel-wise median.

Why We Still Use L7



L7 bridges the gap between L5 (ended 2012) and L8 (started 2013). Without L7, we would lose data continuity in the 2003–2013 period. More data = better composites, even with stripes.

Lab 1: 26-Year NDVI Time Series

40 min hands-on

L5+L7+L8+L9 → Harmonize → Cloud Mask → Annual NDVI → Trend Plot

1

Load 4 Landsat

L5 TM + L7 ETM+
L8 OLI + L9 OLI-2
Filter by AOI & date

2

Harmonize Bands

rename_l57()
rename_l89()
Common band names

3

Scale + Mask

Apply scale factors
 $0.0000275 + (-0.2)$
QA_PIXEL cloud mask

4

Annual NDVI

For each year:
Median composite
Compute NDVI

5

Trend Chart

26-year bar chart
Mark earthquake year
Identify patterns

Expected Output

26-year bar chart of annual median NDVI (2000–2026)
Vertical dashed line at 2024 marking the earthquake year
Long-term baseline vs short-term disruption clearly visible
Statistics panel: mean, std, min year, max year
Key question: Is 2024 an outlier, or part of a longer trend?

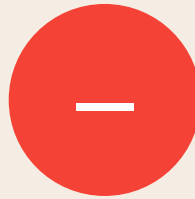
Pixel-Level Linear Regression

$$\text{NDVI}(t) = \text{slope} \times t + \text{offset}$$

`ee.Reducer.linearFit()` computes slope and offset for every pixel across 26 years of annual NDVI



Slope > 0 (Greening)
Long-term vegetation increase
Reforestation, succession
Recovery after disturbance



Slope < 0 (Browning)
Long-term vegetation decline
Urbanization, deforestation
Repeated disturbance



Slope ≈ 0 (Stable)
No significant trend
Mature forest, grassland
Established agriculture

W13 linearFit used 6 years of Sentinel-2 data. W14 extends to 26 years of Landsat -- capturing trends invisible in a short time window. A 6-year decline might be seasonal noise; a 26-year decline is a real signal.

NDVI Trend Map: Reading the Landscape

The slope map tells you WHERE the landscape is changing and in WHICH direction over 26 years

River Corridors

Often show browning (negative slope) due to erosion, channel migration, and gravel deposition. MNDWI in Lab 2 will confirm.

Mountain Slopes

Mixed signal: greening on recovering slopes vs browning on actively eroding areas. Earthquake impact visible as recent browning.

Coastal / Urban

Browning from urban expansion. Greening near restored wetlands or managed forests.

Trend Statistics (computed in Demo)

XX%

greening pixels

XX%

browning pixels

XX%

stable pixels

(Live statistics from ee.Reducer -- fill in during demo!)

Color scale: Red (browning, slope < -0.002) → White (stable) → Green (greening, slope > 0.002)

MNDWI: Water Body Detection

$$\text{MNDWI} = (\text{Green} - \text{SWIR1}) / (\text{Green} + \text{SWIR1})$$

Modified Normalized Difference Water Index — superior to NDWI for suppressing built-up areas

MNDWI > 0.1 = Water

Green light reflects strongly from water, while SWIR is absorbed. High MNDWI = water surface confidently detected.

MNDWI < 0 = Land

SWIR reflects from soil and vegetation more than Green. Negative MNDWI = definitely not water.

Application: Taoyuan Pond Disappearance

By computing MNDWI for every year (2000–2026), we create a 26-year record of where ponds existed. This reveals pond disappearance — which ponds have survived, which are shrinking, and which have been filled for development. Taoyuan Plateau once had over 8,000 irrigation ponds — among the densest pond landscapes in the world. Their loss due to urbanization reduces flood retention capacity and increases urban flood risk.

Lab 2: Taoyuan Pond Disappearance

30 min hands-on

MNDWI → Water Mask → Frequency Map → Disappearance Detection

1

Annual MNDWI

Compute MNDWI
for each year
Threshold > 0.1 = water

2

Water Frequency

Fraction of 26 years
each pixel classified
as water (0–1.0)

3

Period Comparison

Early (2000–2005)
vs recent (2021–2026)
water masks

4

Change Map

Lost ponds = urbanization
New water = retention ponds
Area estimation

Expected Output

Water frequency map: blue gradient showing surviving ponds (freq = 1.0) vs shrinking ponds (freq = 0.1–0.5)

Change map: red = lost ponds (filled for development), green = new water (retention ponds)

Statistics: total water area change, new water km², lost water km²

Key insight: how many ponds have disappeared reveals the scale of urbanization pressure on Taoyuan Plateau

Water Frequency Map: Taoyuan Ponds 26 Years

Annual Water Masks

For each year, MNDWI > 0.1
classifies each pixel as
water (1) or land (0)

Sum Over 26 Years

Count how many years
each pixel was classified
as water (0–26)

Frequency Map

Divide by 26 to get
fraction (0.0–1.0)
Permanent vs transient water

Interpretation Guide

1.0

Surviving ponds — continuous water for 26 years

0.5–0.9

Shrinking ponds — losing area or seasonally dry

0.1–0.5

Disappearing — early water, now mostly land

< 0.1

Non-water — farmland, buildings, roads

Pond Disappearance Detection



Why This Matters: Urban Flood Risk

Taoyuan's 8,000+ ponds served as natural flood retention for centuries. Urbanization is erasing them.

Every filled pond reduces the area's capacity to absorb heavy rainfall — increasing urban flooding.

The 26-year change map quantifies pond loss — essential data for urban resilience planning.

Taoyuan City now promotes pond conservation as cultural landscapes and flood mitigation infrastructure.

XX

km² lost ponds

XX

km² new water

XX

km² net loss

Vegetation Resilience

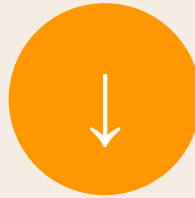
Resilience = the ability of an ecosystem to recover from disturbance

After an earthquake or typhoon, does the vegetation bounce back? How fast? How completely?



High Resilience

NDVI drops sharply, recovers within 1–2 years. Forest regrowth, revegetation.



Low Resilience

NDVI drops and stays low. Repeated disturbance, degraded soil.



Non-Recovery

Permanent land cover change. Landslide scar becomes bare rock.

$$\text{Recovery Ratio} = (\text{Recovery_NDVI} - \text{Impact_NDVI}) / (\text{Baseline_NDVI} - \text{Impact_NDVI})$$

Ratio = 1.0: Full recovery | Ratio = 0.5: Half recovered | Ratio > 1.0: Overshoot | Ratio < 0: Declining

Lab 3: Vegetation Resilience Metrics

20 min hands-on

Baseline → Impact → Recovery → Ratio Map → Statistics

Baseline 2020–2023

Pre-earthquake NDVI
Median of 4 years
Healthy reference state

Impact 2024

Earthquake year NDVI
Immediate damage
Maximum disturbance

Recovery 2025–2026

Post-earthquake NDVI
1–2 years after event
Degree of regrowth

Expected Output

Recovery ratio map: green (ratio > 0.8, strong recovery) to red (ratio < 0.2, poor recovery)

Statistics: % of impacted pixels showing > 50% recovery, mean recovery ratio

Mask: only pixels with significant damage (baseline - impact > 0.1) are evaluated

Key question: does the mountain or the river corridor recover faster? Why?

XX%

strong recovery

XX%

partial recovery

XX%

non-recovery

Cross-Week Integration: Space-Time

W6: Kriging Spatial Gap-Filling

Rain gauges → continuous surface
Spatial interpolation fills
gaps between point data

W13: GEE S2 6-Year Time Series

High resolution (10m)
Short time window
"What happened after EQ?"

W14: GEE Landsat 26-Year Trends

Medium resolution (30m)
Long time window
"What has changed for decades?"

Complete Space-Time Picture of Environmental Change

Spatial interpolation + temporal trend analysis = comprehensive understanding

W6 fills spatial gaps (between rain gauges → continuous rainfall surface).

W14 fills temporal gaps (between snapshots → continuous trend over decades).

Together: a 4D view (x, y, time, value) of how Taiwan's landscapes are transforming.

ARIA v9.5 Architecture

Cloud Layer: GEE + Landsat Multi-Mission Archive

L5+L7+L8+L9

Harmonize

NDVI / MNDWI

linearFit

Resilience

Export

▼ GeoTIFF ▼

Local Layer: Python / QGIS (W8–W12 Tools)

rasterio

scikit-learn

Random Forest

Change Det.

QGIS

ARIA Version History

v5.0

W8: First Light

v6.0

W9: Change Det.

v7.0

W10: SAR Engine

v7.5

W11: Terrain

v8.0

W12: ML Classifier

v9.0

W13: Cloud Engine

v9.5

W14: Resilience

From Snapshots to Decades

Reflection

1

Time Reveals Truth

A 6-year decline might be noise. A 26-year decline is a signal. Longer time series separate real trends from short-term variability.

2

Rivers Are Dynamic

MNDWI over 26 years shows that rivers migrate, erode banks, and reshape landscapes. Static maps miss this fundamental process.

3

Resilience Is Measurable

We can quantify how fast ecosystems recover from disturbance. This transforms disaster response from reactive to predictive.